



Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 24 Further Statistics 2

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General Comments

This was an accessible paper, and learners were able to make progress across all four questions. Very few parts were left blank, and many learners continued to work through the longer questions to completion. Overall, responses suggested that learners were well prepared for this paper.

Comments on Individual Questions

Question 1

Part (a)

This was a standard calculation of Spearman's rank correlation and was answered well by almost all the learners.

Part (b)

Most learners were able to carry out a suitable hypothesis test for Jack's belief. Responses usually stated the hypotheses in terms of ρ , rather than in words, and identified that a two-tailed test was required. Some responses used the incorrect column in the tables when finding the critical value. In some cases, the conclusion was not consistent with the test result, with learners stating that their value of r_s supported Jack's belief.

Question 2

Part (a)

This part was answered well, with most responses showing a single straight line with a positive gradient and labelling a and b correctly. Some responses drew a line that suggested $f(x)$ could be less than 0 for some values.

Part (b)

Most responses identified b correctly, but some did not use the range to find a . Some learners who did not find numerical values for a and b in part (b) were still able to gain method marks in parts (c) and (d).

Part (c)

Most responses used integration, rather than the area of a trapezium, to find the value of k . The integration was usually carried out accurately, and a correct

equation was then solved to reach the given answer. Verification by substitution was not commonly seen.

Part (d)

The common approach was to integrate using limits of -0.5 and x though some responses used an indefinite integral and then used the fact that $F\left(\frac{7}{6}\right) = 1$.

Some responses simply integrated the given probability density function without the “+c” but this incorrect version for $F(x)$ could receive a mark in part (e) but no further marks here.

Most responses reached the correct multiplied out form for $F(x)$ and usually gave a fully correct representation for all values of x .

Part (e)

The majority learners knew how to use their $F(x)$ to find the lower quartile and many succeeded in forming a correct quadratic equation which was usually solved on their calculators and the correct root identified.

Question 3

Part (a)

Learners needed to identify the dependent variable in order to use the formulae in the formula booklet. Most responses did this correctly, and this part was answered very well. A small number of responses gave answers to only 2 significant figures.

Part (b)

The calculation of r was usually carried out accurately. There was less success with part (ii), where some responses explained that the result was consistent because the value of r was close to 1. Other responses stated only that there was “strong correlation”, which was not sufficient. For a sample size of 19, a value of r around 0.6 may still indicate “strong correlation”, while the points may not lie close to a straight line.

Part (c)

The calculation of the residual was answered well with the majority knowing to find the difference between the observed value and the estimated value and most getting the sign correct.

Part (d)

This part was challenging for many learners. Many responses calculated the sum of the residuals for females, noted that it was negative, and in some cases commented that the sum should be zero. Some responses observed that most

of the residuals for females were negative. Fewer responses went on to explain that this indicated the model was, in general, overestimating the metabolic rates of females.

Question 4

Part (a)

Most learners attempted this part. Around half of the responses used the formulae for the mean and variance of a uniform distribution given in the formula booklet. This enabled them to find an expression in c for $E(X^2)$, substitute for $E(T)$, and obtain an equation for c which was usually successfully simplified to the given form.

The other half chose to calculate $E(T)$ directly by using the appropriate integral, $\int_c^4 (x - 1)^2 \frac{1}{4 - c} dx$. These responses often made a good start by integrating and substituting the limits. Some responses were less successful once they reached the resulting cubic equation $c^3 - 3c^2 - 60c + 224 = 0$. Only a few managed to take out the factor $(c - 4)$ and therefore complete the proof or find the roots of the cubic equation, dismiss 4 and 7 and then show that -8 did indeed satisfy the given equation to achieve the final mark.

Part (b)

Even those responses who could not complete the proof in part (a) were able to use the given equation, find the value of c and use it successfully to find the required probability. This part was answered very well, with only a small number of responses not gaining marks for minor errors.